# Truth Tables for Combinations

A washing machine checks three things before it spins.




National 5 Engineering Science

# Daily Review



1. **AND**: A = 1, B = 0. **State** Z.
2. **State** what the **bubble** on a NOT gate shows.
3. **State** the logic state of a **lamp** that is **on**.
4. **State** what **pinMode** sets.
5. **State** what happens to **R** of a **thermistor** when hot.

PART 1

# Rows and Two Inputs

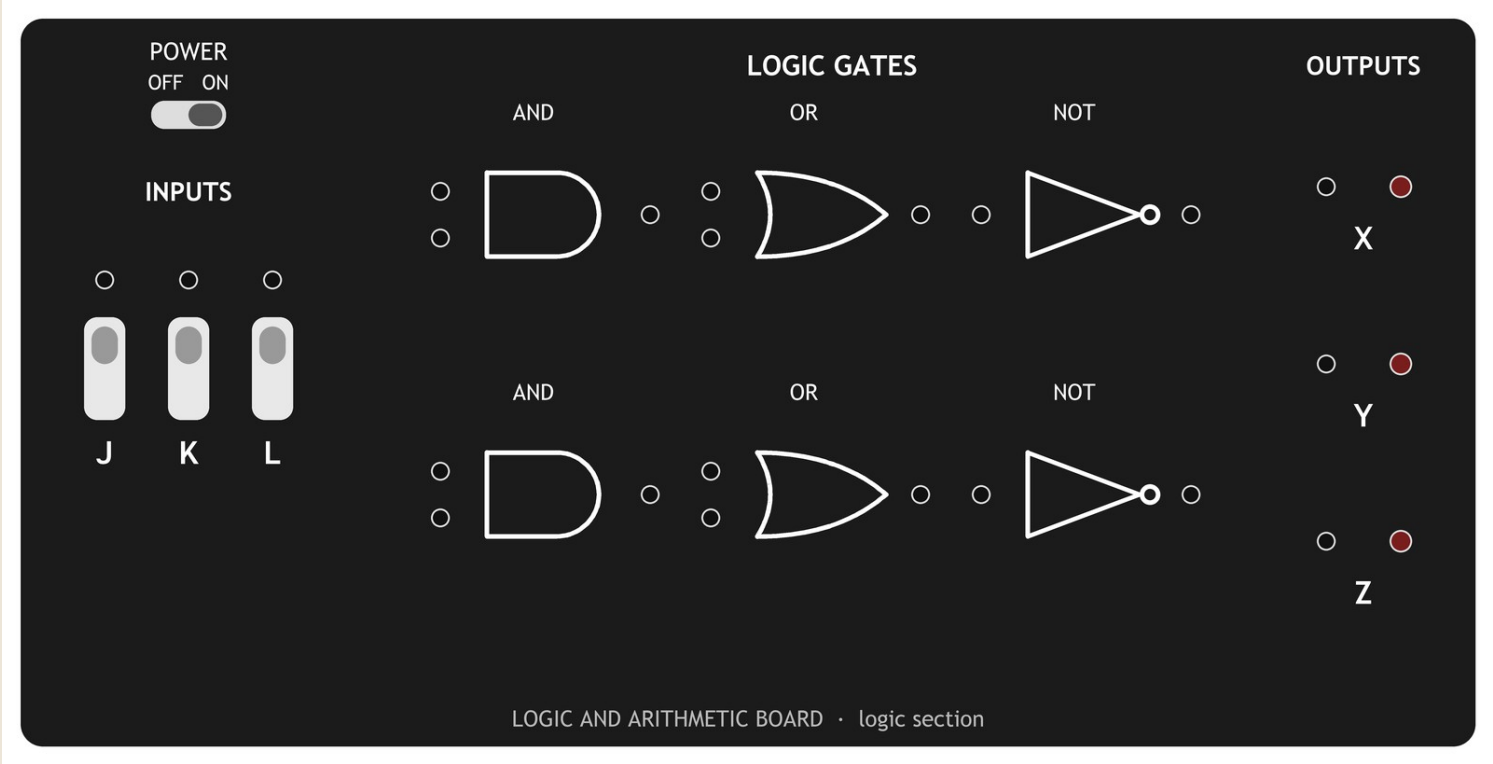
We are learning to count truth table rows and to work out two-input combinations one gate at a time.

# By the end of this part…



1. I can **state** how many **rows** a truth table needs
2. I can **fill in** the **input columns** in order
3. I can **complete** a truth table **one gate at a time**

# Three switches. How many different ways can they be set?



# How Many Rows?

Rows = $2^n$

n = number of **inputs**.

1 → **2**, 2 → **4**, 3 → **8**.

| A |
|---|
| 0 |
| 1 |

| A | B |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |

| A | B | C |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |
| 1 | 1 | 1 |

# Filling the Input Columns

**C** column: **0 1 0 1…**

**B** column: **0 0 1 1…**

**A** column: **0 0 0 0 1 1 1 1.**

Counting in **binary**, so **no row** is missed.

| A | B | C |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |
| 1 | 1 | 1 |

# Show Me: Rows




1. **State** how many rows for **1** input.
2. **State** how many rows for **3** inputs.
3. **State** the **4th** row for 3 inputs.
4. **State** the row after **1 0 1**.
5. **State** the **last** row for 2 inputs.

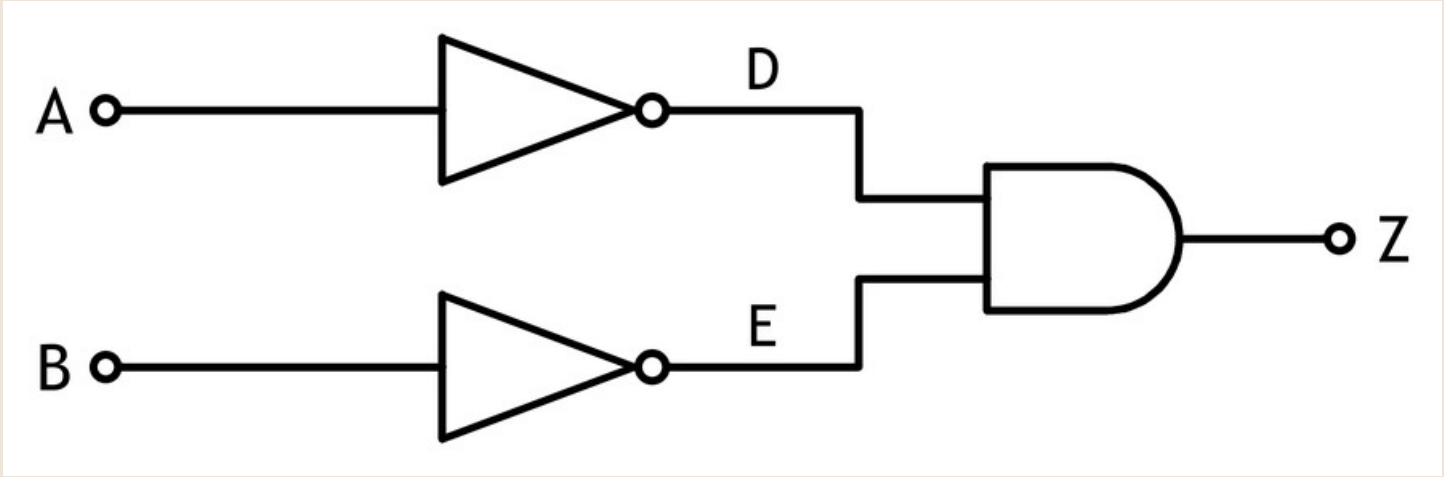
# One Gate at a Time

**Combinational**: no memory.

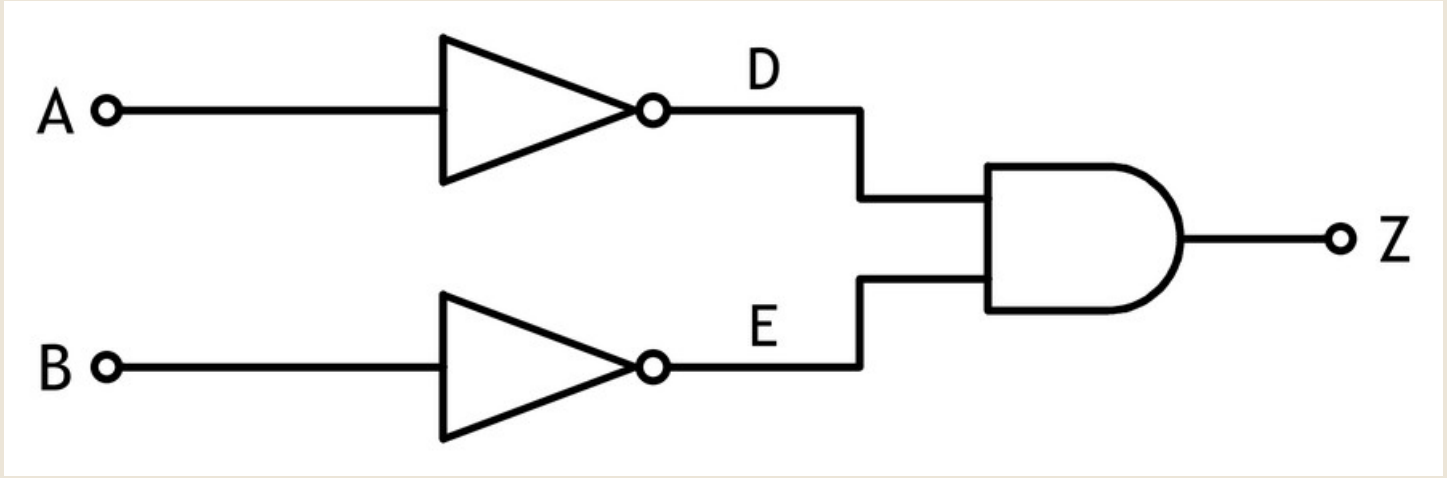
**Label** each gate's output: **D**, **E**.

One **column** each, **Z** last.

Your booklet calls it **X**.



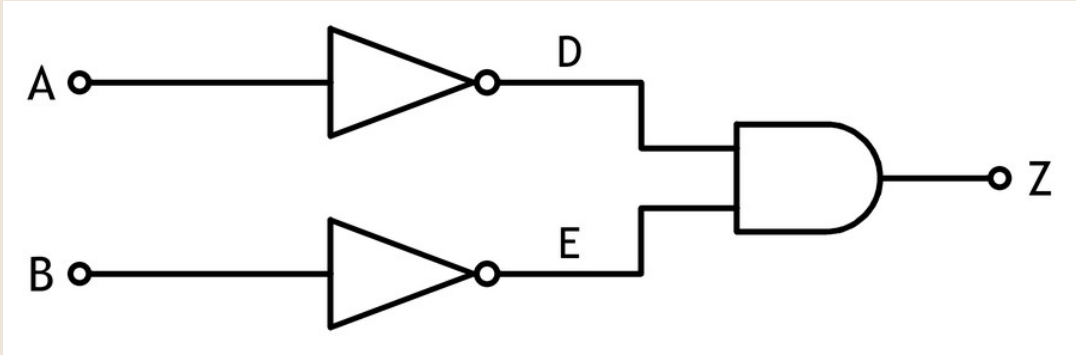
# Worked Example 1




| A | B | D | E | Z |
|---|---|---|---|---|
| 0 | 0 | | | |
| 0 | 1 | | | |
| 1 | 0 | | | |
| 1 | 1 | | | |

Complete the truth table for the logic diagram shown above. **3**

# Worked Example 1: Marking




Complete the truth table for the logic diagram shown above. **3**

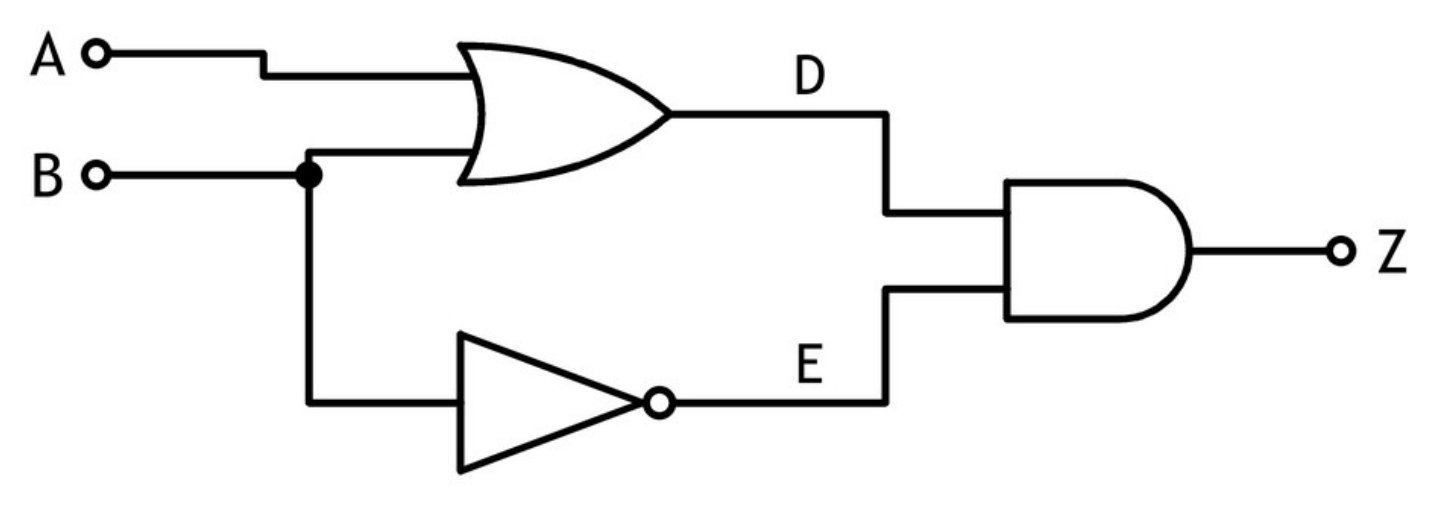
1 mark: column **D** = NOT A

1 mark: column **E** = NOT B

1 mark: column **Z** = D AND E

| A | B | D | E | Z |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 |

# Show Me: Your Turn




Complete the truth table for the logic diagram shown above.

**3**

| A | B | D | E | Z |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 |

# Task 4 (B4a-L3-p11)

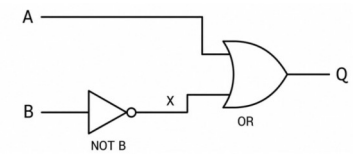
# Task 4

**(a) Complete** the truth table.

**(b) Draw** the diagram, then **complete** the table.

**TASK 4 — Two-Input Truth Tables**

**(a) Q = A OR (NOT B)**




| A | B | X | Q |
|---|---|---|---|
| 0 | 0 | | |
| 0 | 1 | | |
| 1 | 0 | | |
| 1 | 1 | | |

**(b) Q = (NOT A) AND B**

Space for diagram:

| A | B | X | Q |
|---|---|---|---|
| 0 | 0 | | |
| 0 | 1 | | |
| 1 | 0 | | |
| 1 | 1 | | |

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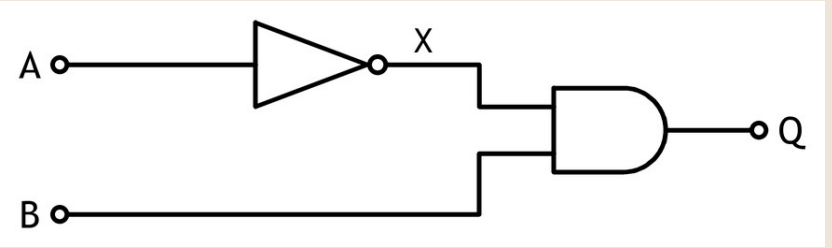
# Answers: Task 4

(a)

| A | B | X | Q |
|---|---|---|---|
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |

(b)

| A | B | X | Q |
|---|---|---|---|
| 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 |



# Look back — by the end of this part…



1. I can **state** how many **rows** a truth table needs
2. I can **fill in** the **input columns** in order
3. I can **complete** a truth table **one gate at a time**

# Daily Review



1. **State** how many rows for **3** inputs.
2. **State** the **3rd** row for 2 inputs.
3. **AND**: A = 1, B = 1. **State** Z.
4. **OR**: A = 0, B = 0. **State** Z.
5. **NOT**: A = 0. **State** Z.

PART 2

# Three Inputs

We are learning to complete a three-input truth table the way the exam marks it.

# By the end of this part...



1. I can **complete** a **three-input** truth table
2. I can **use** the **D** and **E** columns
3. I can **check** each column **scores** a mark

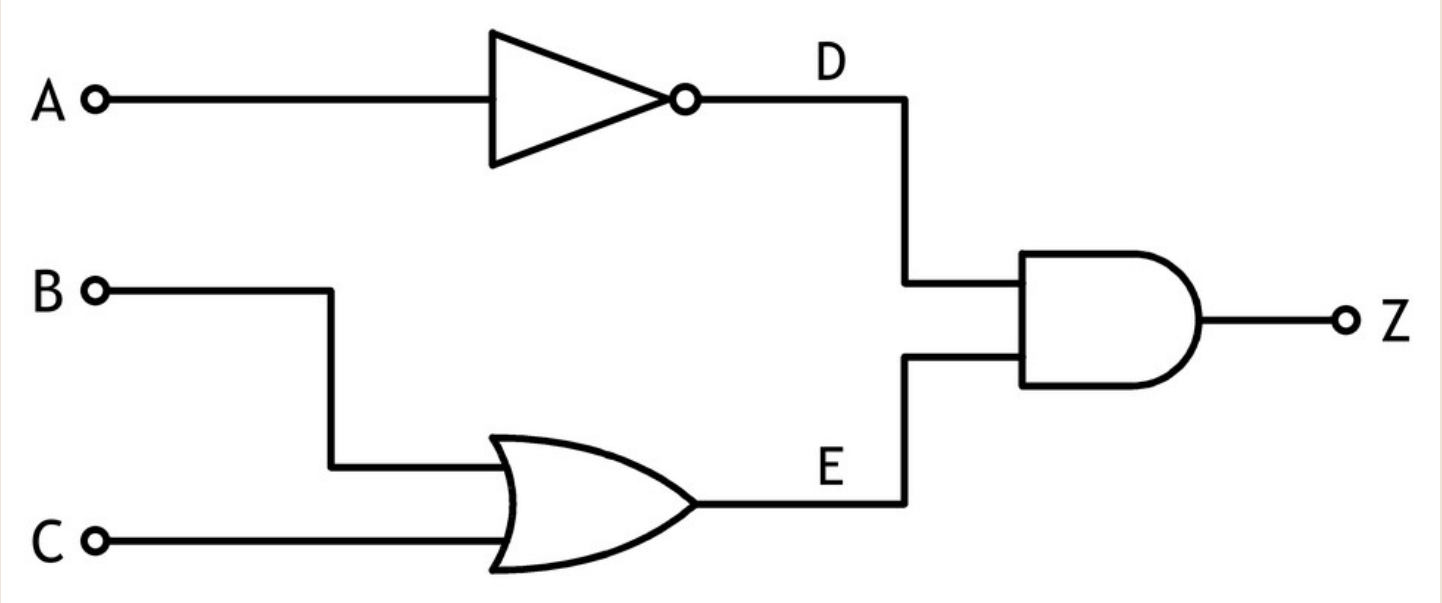
# Three Inputs

**8** rows.

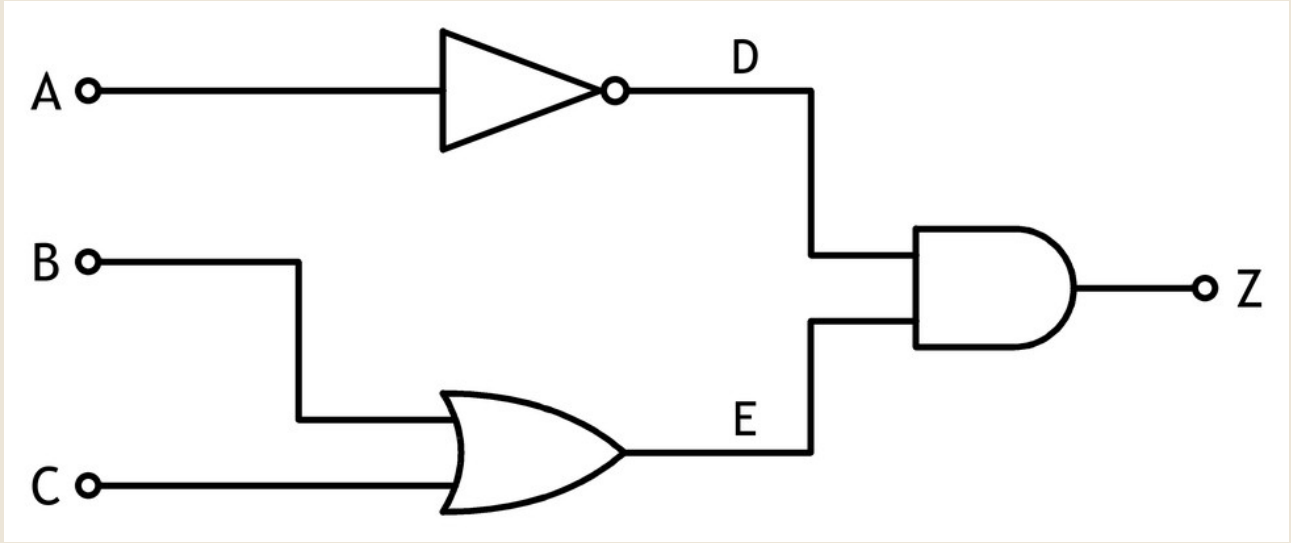
Inputs **A**, **B**, **C**.

Work out **D**, then **E**.

**Z** last, from D and E.



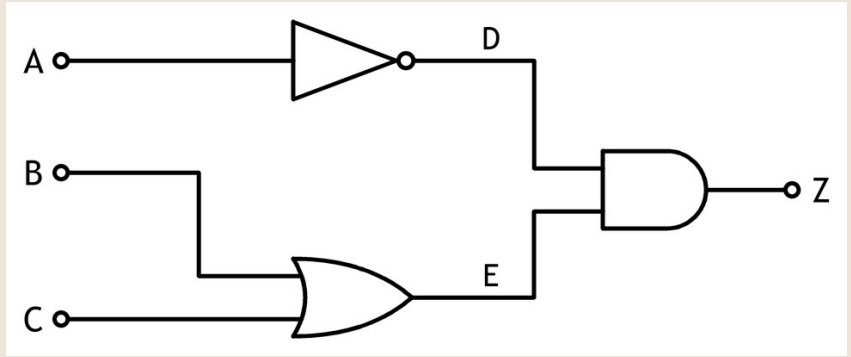
# Worked Example 2




Complete the truth table for the logic diagram shown above. **3**

| A | B | C | D | E | Z |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

# Worked Example 2: Marking




Complete the truth table for the logic diagram shown above. **3**

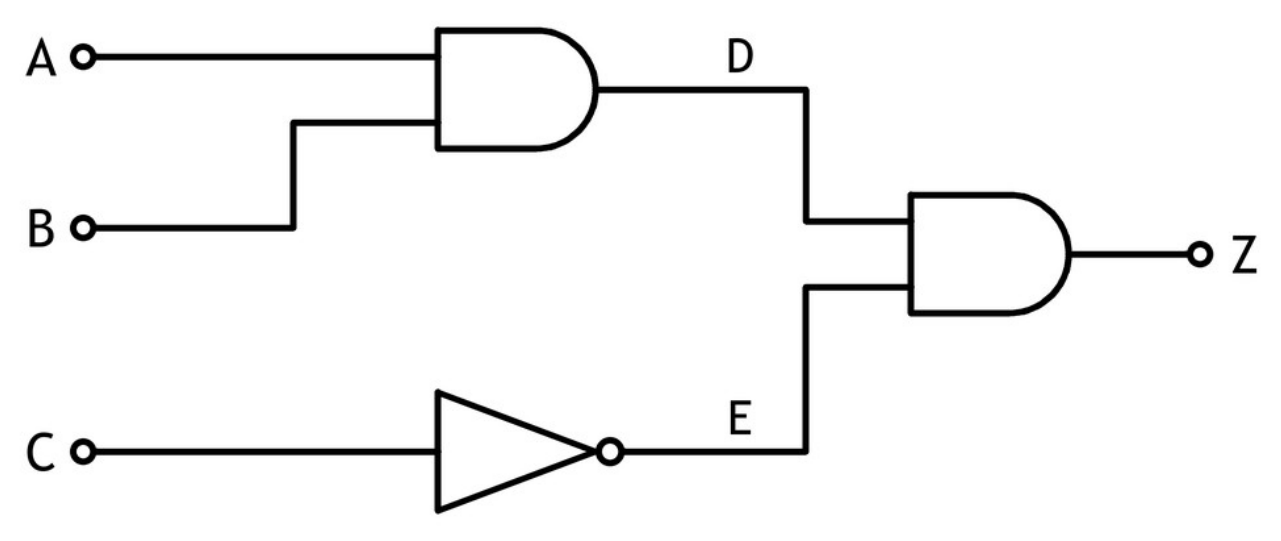
1 mark: **D** = NOT A

1 mark: **E** = B OR C

1 mark: **Z** = D AND E

| A | B | C | D | E | Z |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 1 | 0 |

# Show Me: Your Turn




Complete the truth table for the logic diagram shown above. **3**

| A | B | C | D | E | Z |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 |

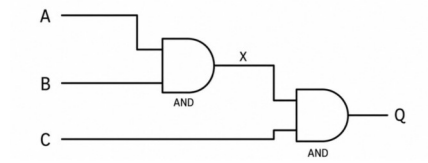
# Task 5 (B4a-L3-p12)

# Task 5

**Complete** the **X** column, then **Q**.

**Three-Input Combinations**

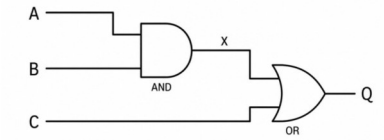
**Worked Example — Q = A AND B AND C**




| A | B | C | X | Q |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 |

**TASK 5 — Three-Input Truth Table**

Complete the table for Q = (A AND B) OR C.




| A | B | C | X | Q |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

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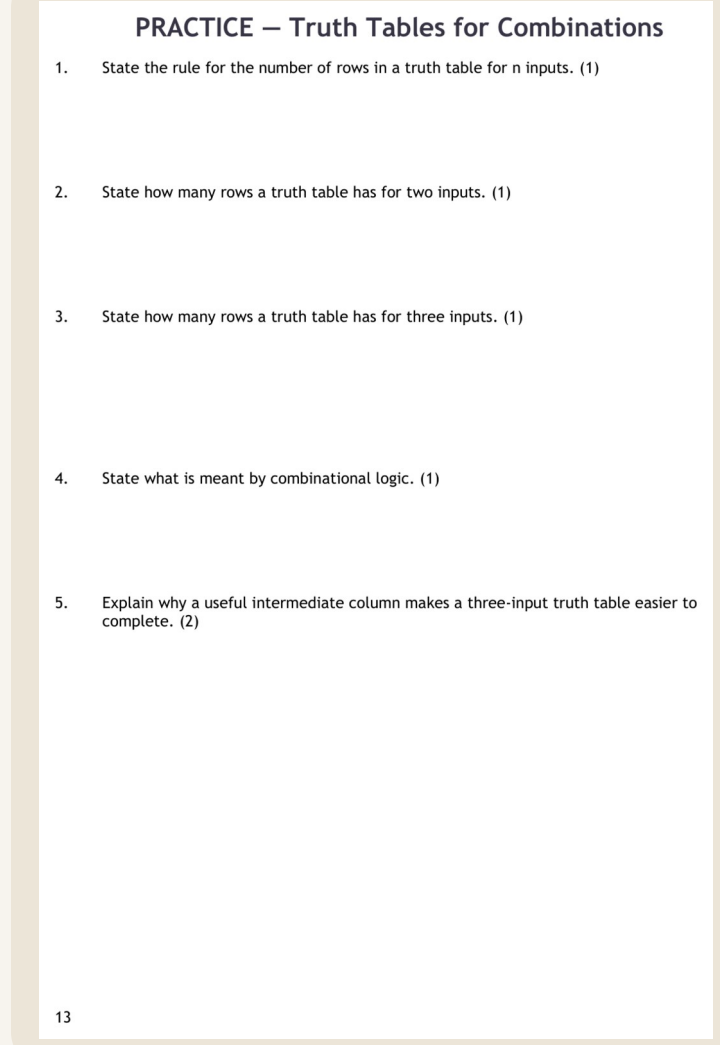
# Answers: Task 5

| A | B | C | X | Q |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |

# **Review Questions** (Booklet page 13)

# Review Questions

**Answer** Questions **1** to **5** on your own.



**PRACTICE — Truth Tables for Combinations**

1. State the rule for the number of rows in a truth table for n inputs. (1)

2. State how many rows a truth table has for two inputs. (1)

3. State how many rows a truth table has for three inputs. (1)

4. State what is meant by combinational logic. (1)

5. Explain why a useful intermediate column makes a three-input truth table easier to complete. (2)

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# Answers: Review Questions

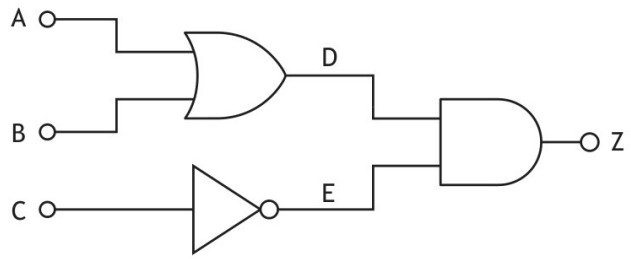


| Q | Mark 1 | Mark 2 |
|---|---|---|
| 1 | Rows = $2^n$ | — |
| 2 | **4** | — |
| 3 | **8** | — |
| 4 | Depends on **inputs now** only | — |
| 5 | One gate **at a time** | **so** mistakes are easy to spot |

# 2025 Section 2 Question 11 (c)

**11. (continued)**

Part of a logic diagram used in the control circuit of the ring light is shown below.




(c) Complete the truth table for the logic diagram shown above.

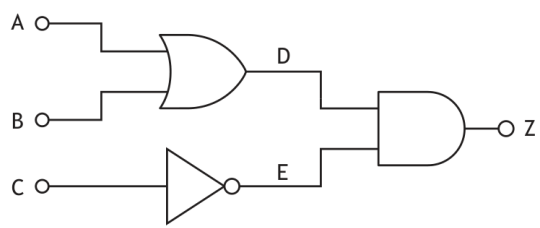
| A | B | C | D | E | Z |
|---|---|---|---|---|---|
| 0 | 0 | 0 | | | |
| 0 | 0 | 1 | | | |
| 0 | 1 | 0 | | | |
| 0 | 1 | 1 | | | |
| 1 | 0 | 0 | | | |
| 1 | 0 | 1 | | | |
| 1 | 1 | 0 | | | |
| 1 | 1 | 1 | | | |

**(c)** is worth
**3 marks.**

**D** first, then **E**, then **Z**.

# Marking: 2025 Section 2 Question 11 (c)

Part of a logic diagram used in the control circuit of the ring light is shown below.




(c) Complete the truth table for the logic diagram shown above.

Complete the truth table for the logic diagram shown above. **3**

1 mark: column **D** = A OR B

1 mark: column **E** = NOT C

1 mark: column **Z** = D AND E

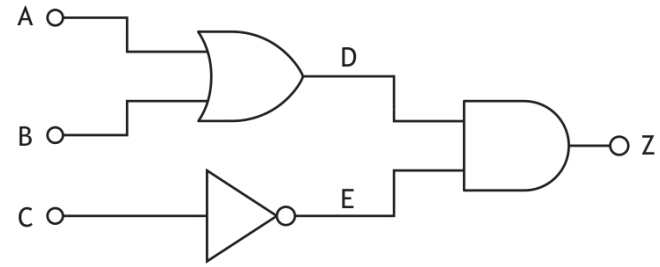
| A | B | C | D | E | Z |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 |

# Truth Tables in the Exam

| Year | Questions | Marks |
|---|---|---|
| 2026 | S2 Q14(d) | 3 |
| 2025 | S2 Q11(c) | 3 |
| 2024 | S2 Q14(c) | 3 |
| 2023 | S2 Q12(f) | 3 |

Kept for the test: 2026 Q14(d), 2024 Q14(c), 2023 Q12(f).

Part of a logic diagram used in the control circuit of the ring light is shown below.




(c) Complete the truth table for the logic diagram shown above.

2025 Q11(c)

# Checkpoint




1. **Explain** why we label **D** and **E**.

2. **State** the rows for **3** inputs.

3. D is wrong. **State** if Z can still score.

# Look back — by the end of this part…



1. I can **complete** a **three-input** truth table
2. I can **use** the **D** and **E** columns
3. I can **check** each column **scores** a mark

# Extension



1. D = NOT A, E = NOT C, Z = D OR E. **Complete** the table.

2. For interest: **state** the rows for **4** inputs.

# Course Specification — 1 of 1

## Digital electronic control systems

- AND, OR and NOT gates and combinations with up to three inputs, using truth tables, logic diagrams and Boolean expressions

# Credits

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R Stewart - November 2026